Introduction to generalized additive models

day 2

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Overview

I. Example 1: Diabetic retinopathy

II. Example 2: Methylation Sequencing

III. Example 3: Primary biliary cirrhosis survival analysis

IV. Example 4: Fuel efficiency of cars

Example 1: Diabetic retinopathy

Example 1: Diabetic retinopathy

Clinical data on whether diabetic patients suffer from diabetic retinopathy or not.

Following information was collected:

- `ret`: binary indicator of retinopathy progression at first follow-up
- `dur`: duration of diabetes at baseline, in years
- `gly`: percent of glycosylated hemoglobin at baseline.
- `bmi`: body mass index at baseline.

The relationship of risk of retinopathy progession to the covariates is unclear.

Example 1: Diabetic retinopathy

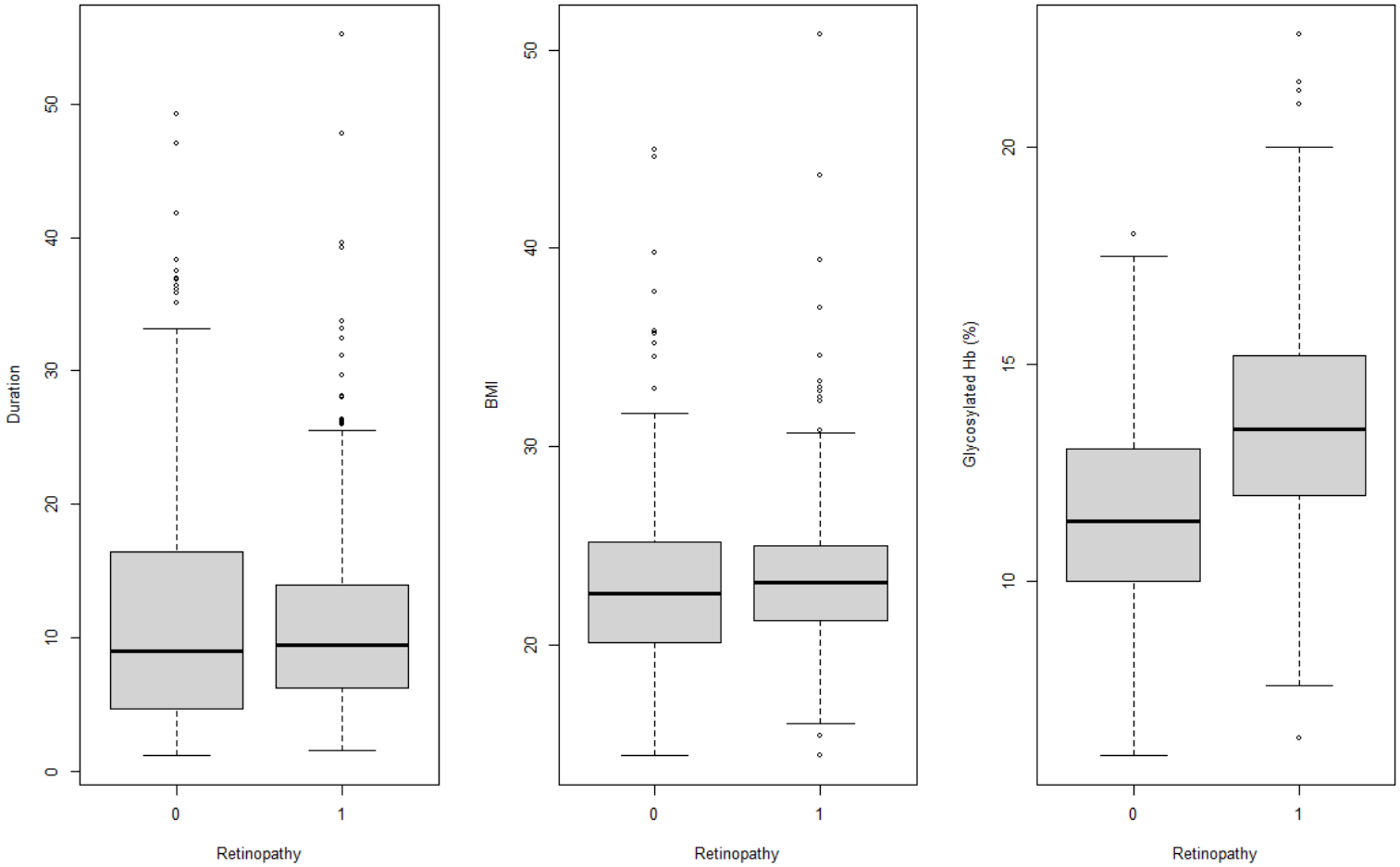


Figure: Boxplots of disease duration, percent glycosylated hemoglobin and body mass index, by retinopathy status

Example 1: Diabetic retinopathy

The relationship of risk of retinopathy progession to the covariates is unclear.

Use an additive logistic regression model to study the link between retinopathy and the explanatory variables:

$$\begin{aligned} \text{logit}(\pi_i) &= f_1(dur_i) + f_2(gly_i) + f_3(bmi_i) + f_4(dur_i, gly_i) \\ &+ f_5(dur_i, bmi_i) + f_6(bmi_i, gly_i) \end{aligned}$$

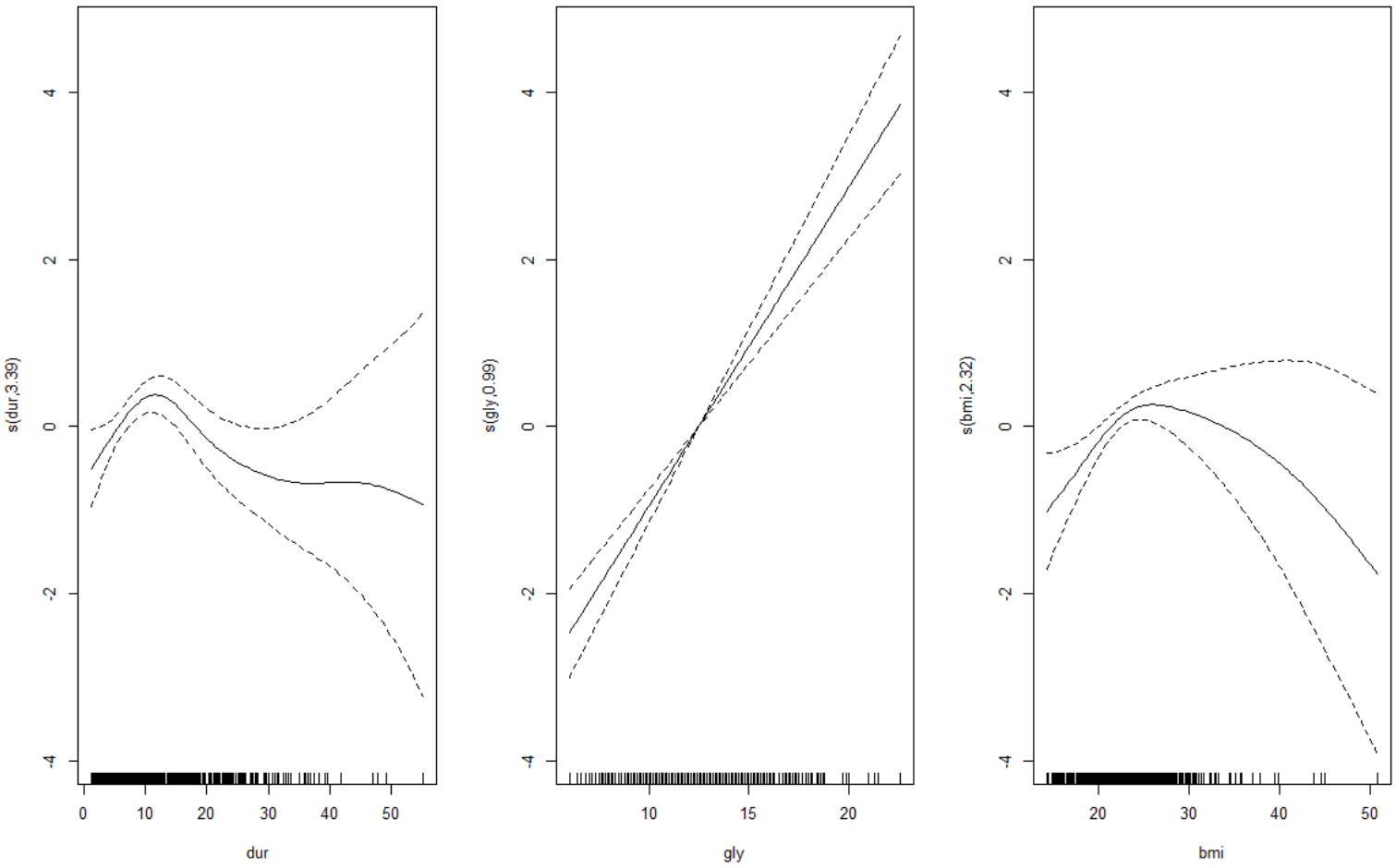
Example 1: Diabetic retinopathy

```
> k <- 8
> ex1 <- gam(ret ~ s(dur,k=k) + s(gly,k=k) + s(bmi,k=k) +
+          ti(dur,gly,k=k) + ti(dur,bmi,k=k) +
+          ti(gly,bmi,k=k), select=TRUE,
+          data=wesdr, family=binomial(),
+          method="ML")
> ex1
```

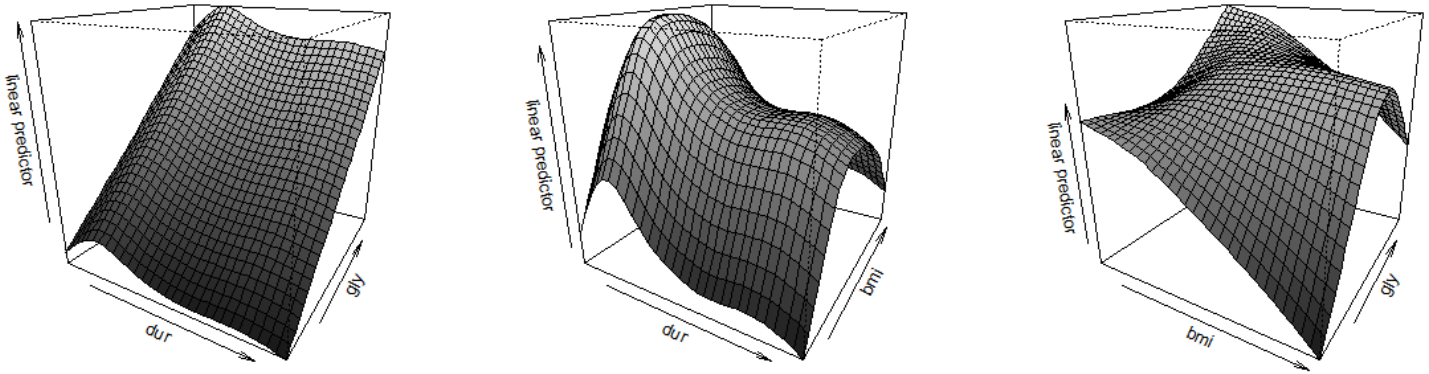
Example 1: Diabetic retinopathy

```
Family: binomial
Link function: logit

Formula:
ret ~ s(dur, k = k) + s(gly, k = k) + s(bmi, k = k)
    + ti(dur, gly, k = k) + ti(dur, bmi, k = k)
    + ti(gly, bmi, k = k)

Estimated degrees of freedom:
3.3926 0.9891 2.3151 0.0004 0.0002 1.6893  total = 9.39
```

Example 1: Diabetic retinopathy



Example 1: Diabetic retinopathy



Example 1: Diabetic retinopathy

Conclusions:

- $f_2(gly_i)$: EDF $\approx 1 \rightarrow$ straightline, no smoother needed
- $f_4(dur_i, gly_i)$: EDF $\approx 0 \rightarrow$ no additional information from interaction terms, main effects suffice
- $f_5(dur_i, gly_i)$: EDF$\approx 0 \rightarrow$ main effects suffice
- $f_6(dur_i, gly_i)$: EDF $> 1 \rightarrow$ main effect and interaction effect are needed

Example 1: Diabetic retinopathy

An updated model could thus be:

$$\text{logit}(\pi_i) = f_1(dur_i) + f_2(bmi_i, gly_i)$$

with $f_2(bmi_i, gly_i)$ a tensor product smoother $\rightarrow$ no longer ANOVA decomposition

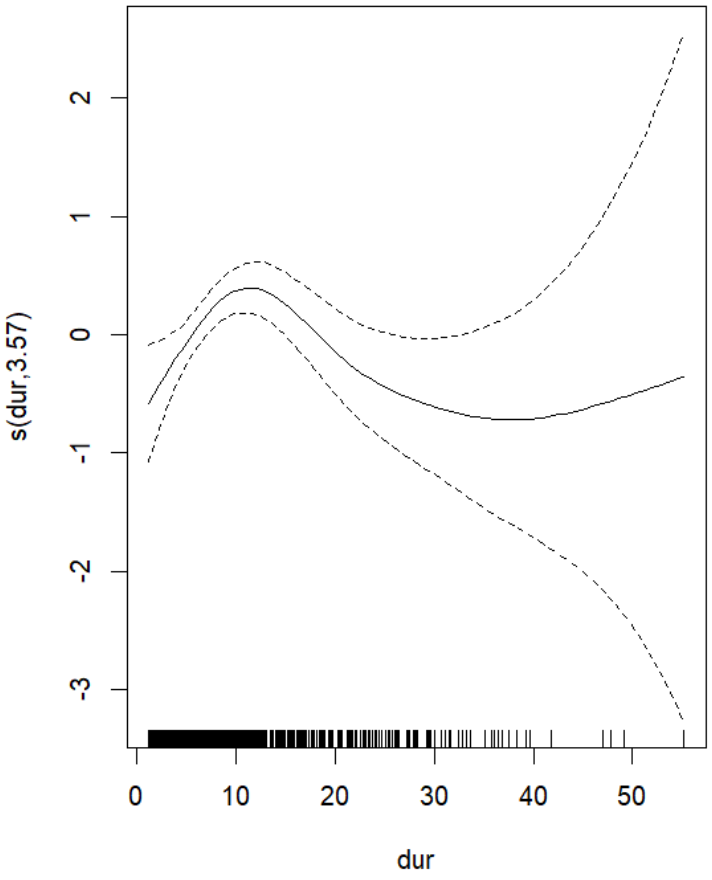
Example 1: Diabetic retinopathy

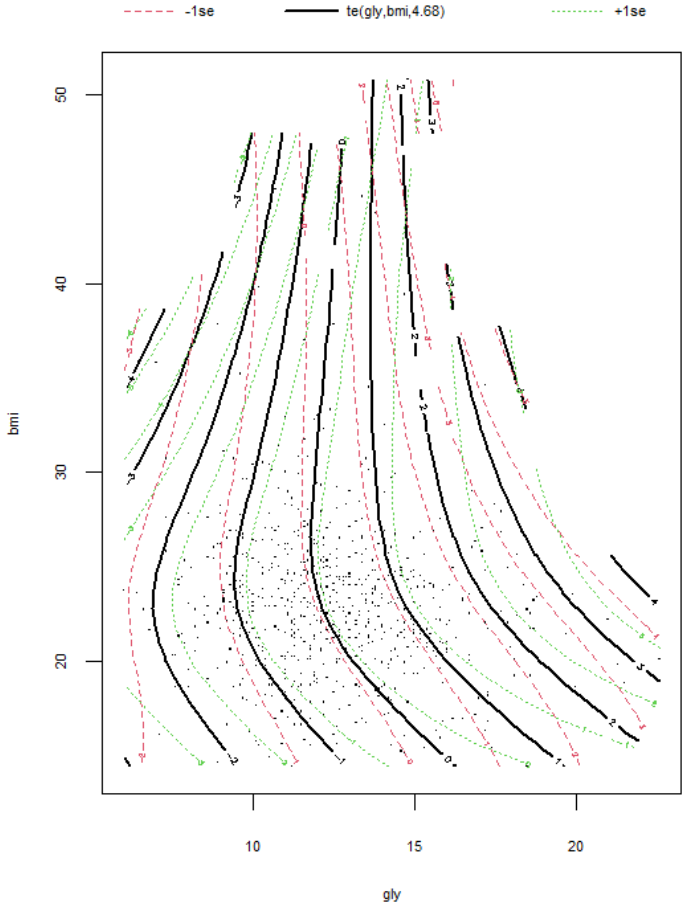
```
> ex1b <- gam(ret ~ s(dur,k=k) + te(gly,bmi,k=k),
+ data=wesdr, family=binomial(), method="ML")
> ex1b
```

Example 1: Diabetic retinopathy

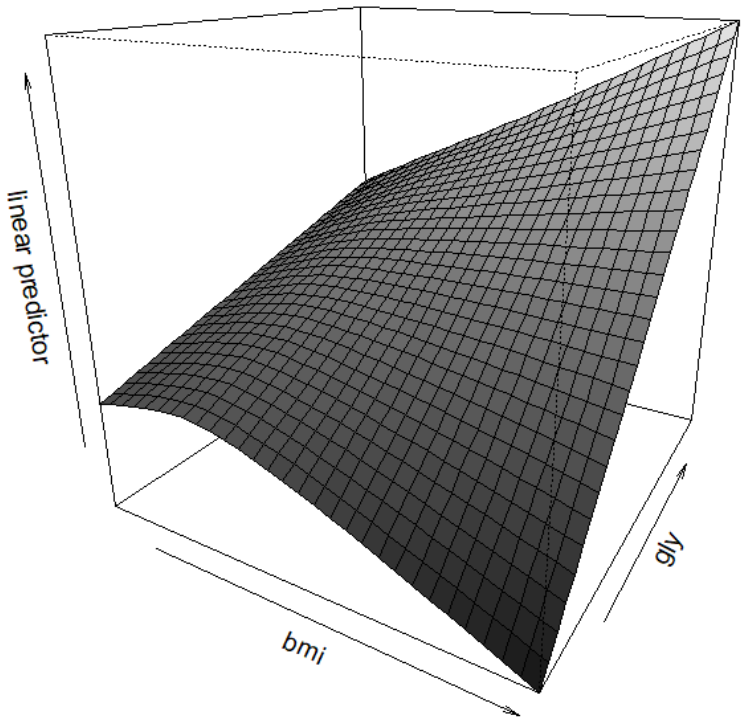
```
Family: binomial
Link function: logit

Formula:
ret ~ s(dur, k = k) + te(gly, bmi, k = k)

Estimated degrees of freedom:
3.57 4.68  total = 9.25
```

Example 1: Diabetic retinopathy





Example 1: Diabetic retinopathy



Example 1: Diabetic retinopathy

```
Approximate significance of smooth terms:
              edf Ref.df Chi.sq p-value
s(dur)      3.572  4.402  16.04 0.00456 **
te(gly,bmi) 4.680  5.588  96.25 < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

R-sq.(adj) =  0.218   Deviance explained =   18%
-ML = 382.21  Scale est. = 1         n = 669
```

Example 2: Methylation Sequencing

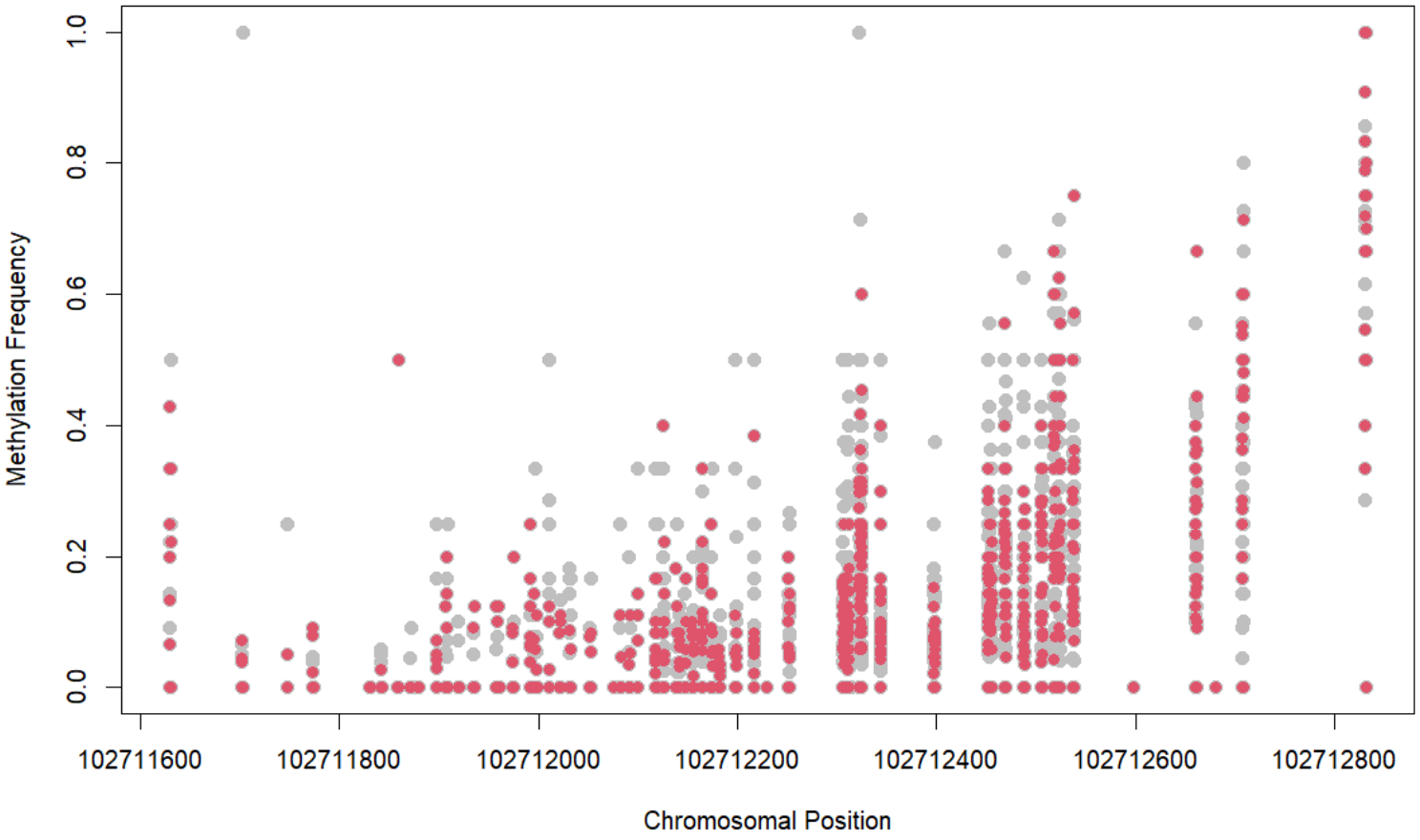
Example2: Methylation Sequencing

Detect differentially methylated regions on a part of Chromosome 4 between patients with reumatoid artritis and patients without reumatoid artritis

Following information was collected:

- `RA`: binary indicator if the person has reumatoid artritis
- `Position`: chromosomal position of a methylation site
- `Total_Counts`: number of times a methylation site has been sequenced (per patient)
- `Meth_Counts`: number of times a methylation has been detected (per patient)
- `T_cell`: binary indicator of the studied cell was the a T-cell or a monocyte

Example 2: Methylation Sequencing



Example 2: Methylation Sequencing

Instead of fitting a generalized additive model, we will use a varying-coefficient model.

Varying coefficient models are an extension of additive models where you have an interaction between a parametric term and a smooth term.

$$\begin{aligned} \text{logit}(\pi_i) &= f_1(pos_i) + f_2(pos_i) \times \text{RA} + f_3(pos_i) \times \text{T-cell} \\ &+ f_4(pos_i) \times (\text{RA} \times \text{T-cell}) \end{aligned}$$

with $Y_i \sim \text{Beta-Binomial}(\pi_i, n_i, \sigma_i)$

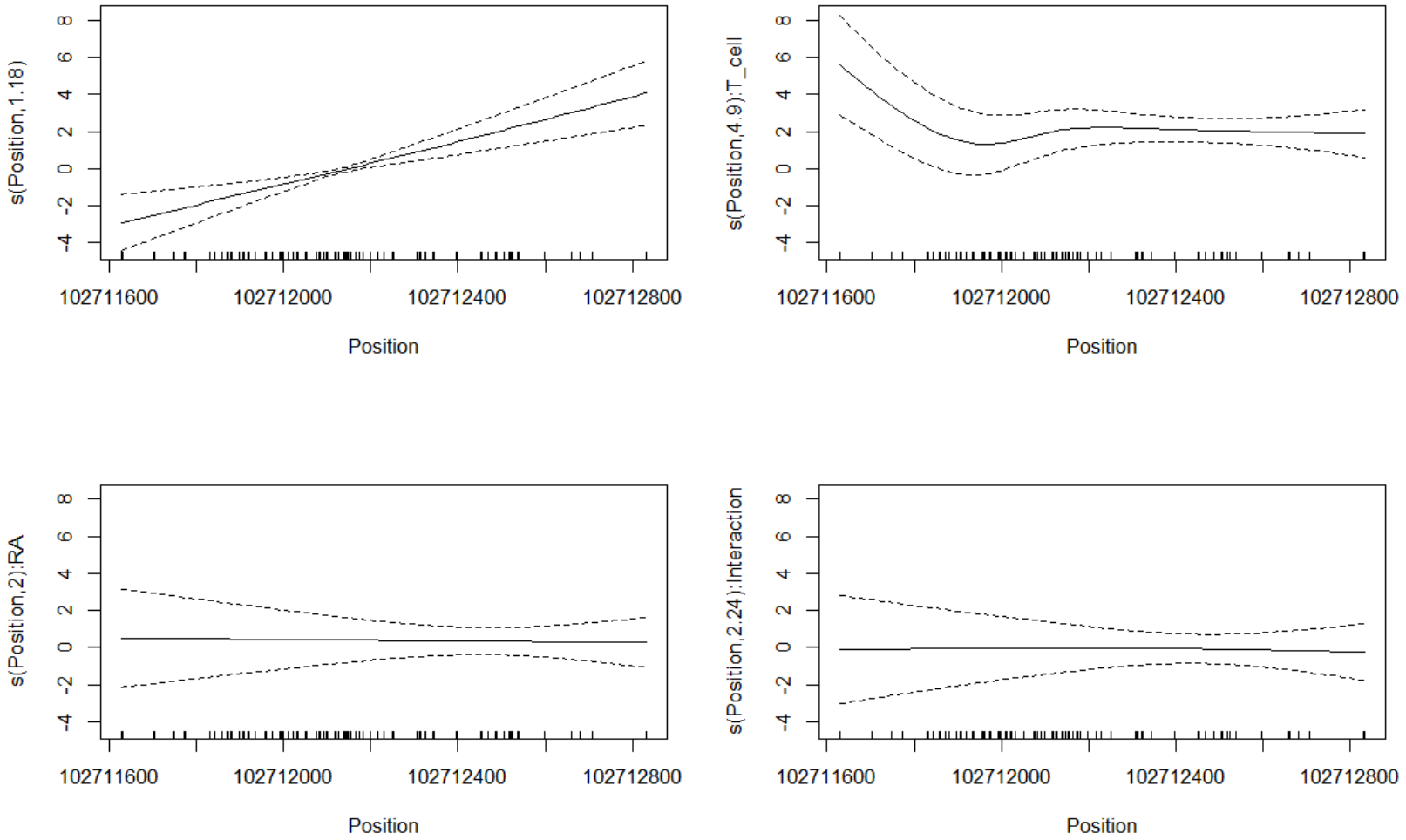
Example 2: Methylation Sequencing

```
> library(gamlss)
> library(gamlss.dist)
> library(gamlss.add)
> ex2a <- gamlss(cbind(Meth_Counts,
+  Total_Counts-Meth_Counts)~ga(
+  ~s(Position, bs="cr", k=5) +
+   s(Position, by=T_cell, bs="cr", k=5) +
+   s(Position, by=RA, bs="cr", k=5) +
+   s(Position, by=Interaction, bs="cr", k=5)),
+  data=dat , family=BB)
> getSmo(ex2a)
```

Example 2: Methylation Sequencing

```
Family: gaussian
Link function: identity

Formula:
Y.var ~ s(Position, bs = "cr", fx = FALSE, k = 5) + ...

Estimated degrees of freedom:
1.18 4.90 2.00 2.24  total = 11.33

GCV score: 7.573957
```

Example 2: Methylation Sequencing



Example 2: Methylation Sequencing

Conclusions:

- based on EDF: all smoothers are fine, perhaps $f_1(pos_i)$
- based on figure: what about significance of $f_3(pos_i) \times$ RA and $f_4(pos_i) \times$ Interaction?

Example 2: Methylation Sequencing

```
Approximate significance of smooth terms:
                          edf Ref.df      F  p-value
s(Position)             1.184  1.337 12.467 7.49e-05 ***
s(Position):T_cell      4.903  4.989 14.831  < 2e-16 ***
s(Position):RA          2.000  2.000  0.442    0.643
s(Position):Interaction 2.244  2.445  0.048    0.958
```

Example 2: Methylation Sequencing

Fit an updated model without interaction term
$f_4(pos_i) \times \text{RA} \times \text{T-cell}$

$$\text{logit}(\pi_i) \quad = \quad f_1(pos_i) + f_2(pos_i) \times \text{RA} + f_3(pos_i) \times \text{T-cell}$$

with $Y_i \sim \text{Beta-Binomial}(\pi_i, n_i, \sigma_i)$

Example 2: Methylation Sequencing

```
> ex2b <- gamlss(cbind(Meth_Counts,
+  Total_Counts-Meth_Counts)~ga(
+  ~s(Position, bs="cr", k=5) +
+    s(Position, by=T_cell, bs="cr",k=5) +
+    s(Position, by=RA, bs="cr",k=5)),
+  data=dat , family=BB)
> getSmo(ex2b)
> summary(ex2b)
```

Example 2: Methylation Sequencing

```
Family: gaussian
Link function: identity

Formula:
Y.var ~ s(Position, bs = "cr", fx = FALSE, k = 5) + ...

Estimated degrees of freedom:
1.04 4.89 2.00  total = 8.93

GCV score: 7.564041
```

Example 2: Methylation Sequencing

```
Approximate significance of smooth terms:
                      edf Ref.df      F p-value
s(Position)         1.036  1.071 47.019  <2e-16 ***
s(Position):T_cell 4.895  4.991 30.080  <2e-16 ***
s(Position):RA     2.000  2.000  2.225   0.108
```

Example 2: Methylation Sequencing

It is also possible to fit an additive model on the scale parameter σ_i.

Therefore, we will fit following model:

$$\begin{aligned} \text{logit}(\pi_i) &= f_1(pos_i) + f_2(pos_i) \times \text{RA} + f_3(pos_i) \times \text{T-cell} \\ \log\sigma_i &= g_1(pos_i) + g_2(pos_i) \times \text{RA} + g_3(pos_i) \times \text{T-cell} \end{aligned}$$

with $Y_i \sim \text{Beta-Binomial}(\pi_i, n_i, \sigma_i)$

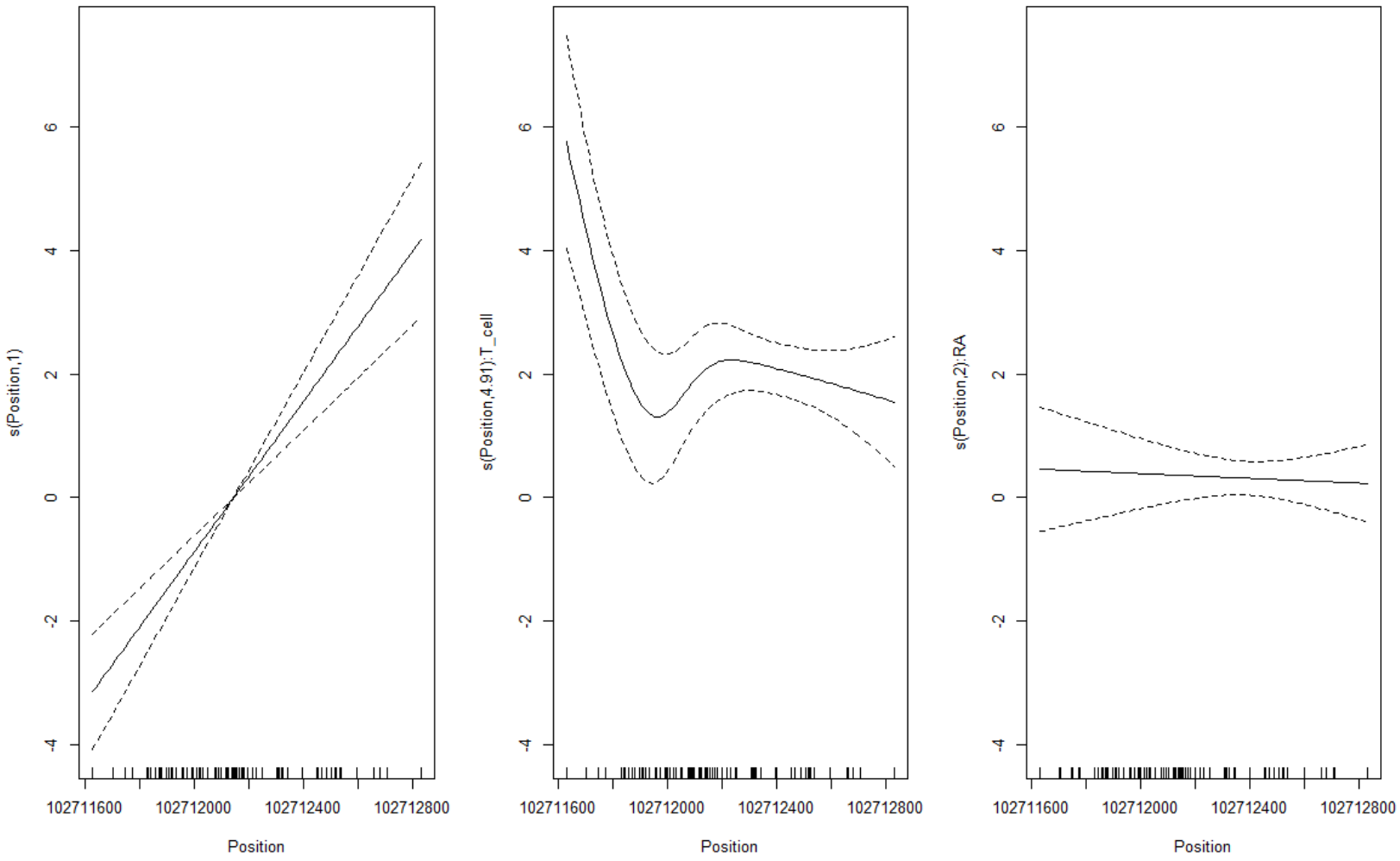
Example 2: Methylation Sequencing

```
> m4 <- gamlss(cbind(Meth_Counts,
+ Total_Counts-Meth_Counts)~ga(
+ ~s(Position, bs="cr" , fx=FALSE, k=5) +
+  s(Position, by=T_cell, bs="cr", fx=FALSE, k=5) +
+  s(Position, by=RA, bs="cr", fx=FALSE, k=5)),
+ sigma.formula=~ga(
+ ~s(Position, bs="cr" , fx=FALSE, k=5) +
+  s(Position, by=T_cell, bs="cr", fx=FALSE, k=5) +
+  s(Position, by=RA, bs="cr", fx=FALSE, k=5)),
+ data=dat , family=BB)
```

Example 2: Methylation Sequencing

```
> plot(getSmo(m4,"mu"))
> (getSmo(m4, "mu"))
> summary(getSmo(m4, "mu"))
```

Example 2: Methylation Sequencing



Example 2: Methylation Sequencing

```
Family: gaussian
Link function: identity

Formula:
Y.var ~ s(Position, bs = "cr", fx = FALSE, k = 5) + ...

Estimated degrees of freedom:
1.00 4.91 2.00  total = 8.91

GCV score: 7.637133
```

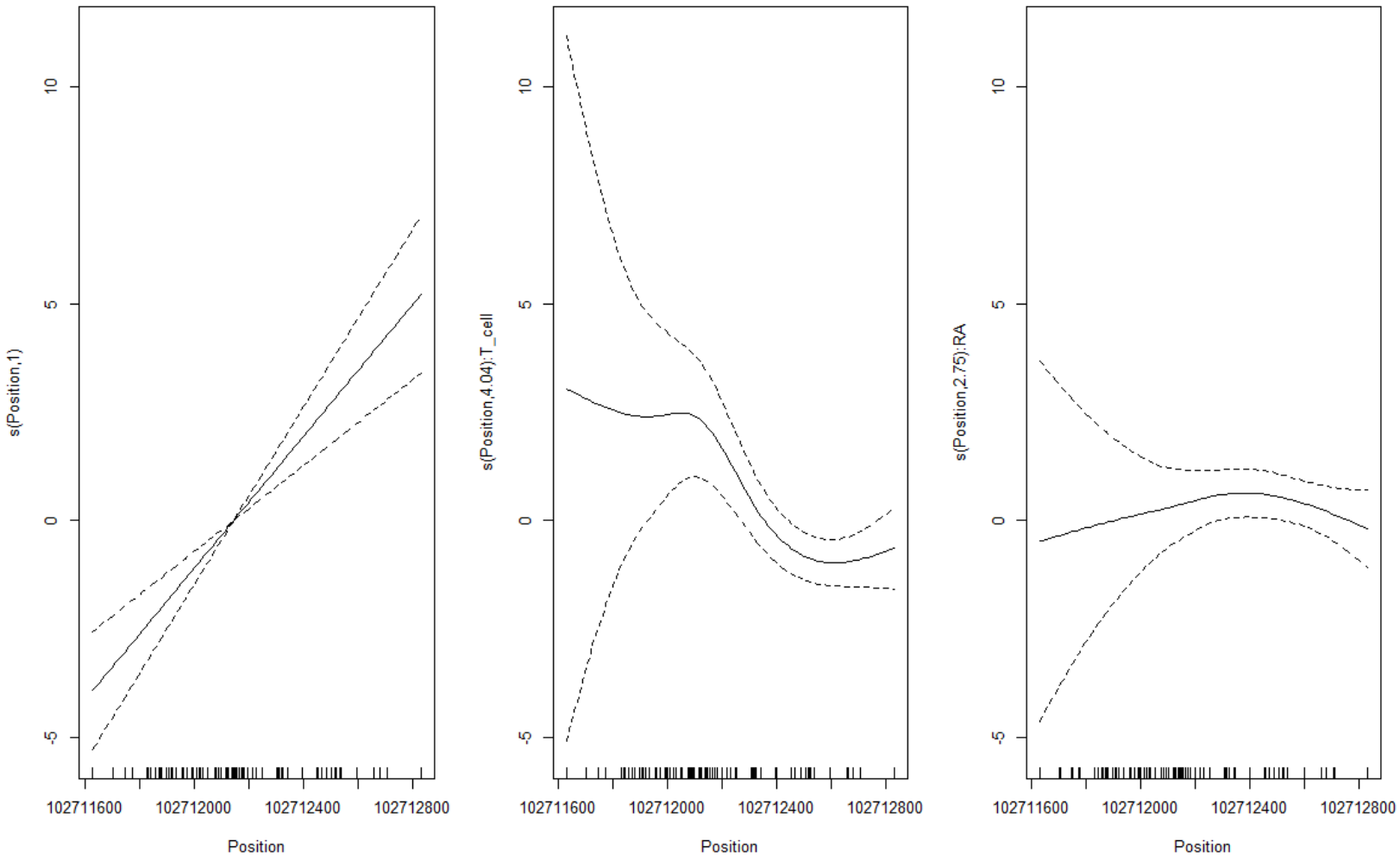
Example 2: Methylation Sequencing

```
Approximate significance of smooth terms:
                      edf Ref.df      F p-value
s(Position)         1.000  1.000 44.738  <2e-16 ***
s(Position):T_cell 4.911  4.994 29.162  <2e-16 ***
s(Position):RA     2.000  2.000  2.707  0.0669 .
```

Example 2: Methylation Sequencing

```
> plot(getSmo(m4,"sigma"))
> (getSmo(m4, "sigma"))
> summary(getSmo(m4, "sigma"))
```

Example 2: Methylation Sequencing



Example 2: Methylation Sequencing

```
Family: gaussian
Link function: identity

Formula:
Y.var ~ s(Position, bs = "cr", fx = FALSE, k = 5) + ...

Estimated degrees of freedom:
1.00 4.04 2.75  total = 8.8

GCV score: 0.9181318
```

Example 2: Methylation Sequencing

```
Approximate significance of smooth terms:
                      edf Ref.df      F  p-value
s(Position)         1.000  1.000 33.091  < 2e-16 ***
s(Position):T_cell 4.044  4.365  6.256 2.43e-05 ***
s(Position):RA     2.753  3.060  1.704    0.159
```

Example 3: Primary biliary cirrhosis survival analysis

Example 3: Primary biliary cirrhosis survival analysis

Primary biliary cirrhosis is a disease caused by autoimmune damage to the bile ducts of the liver. It can eventually lead to cirrhosis of the liver.

Data from a trial conducted at the Mayo clinic between 1974 and 1984 testing the drug D-penicillamine against a placebo

Example 3: Primary biliary cirrhosis survival analysis

Following information was collected during the trial:

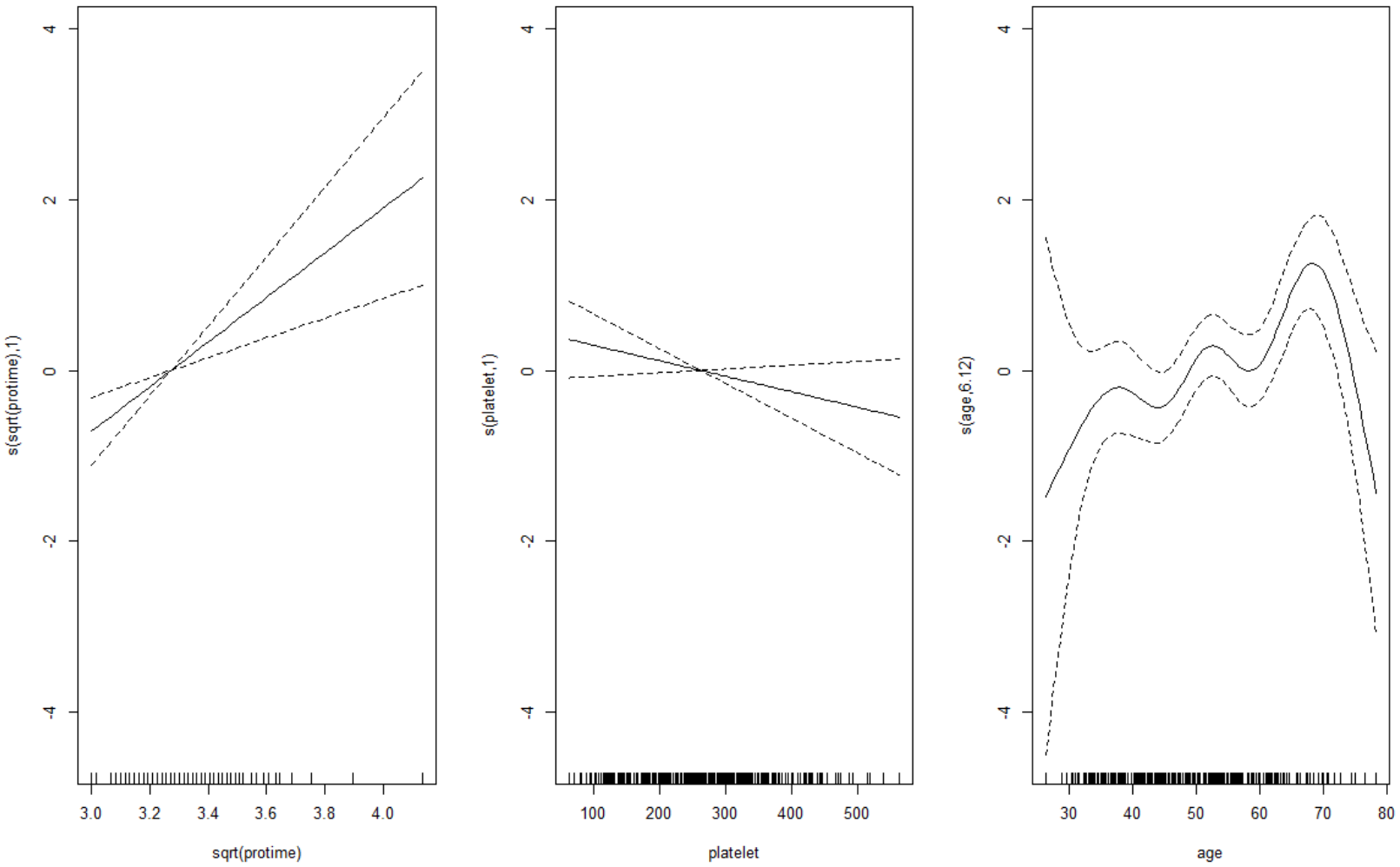
- `age`
- `sex`
- `trt`
- `stage`
- `albumin`
- `alk phosphatase`: alkaline phosphotase
- `ast`: aspartate aminotransferase
- `bili`: bilirubin
- `platelet`: count
- `protime`: standardised blood clotting time

Example 3: Primary biliary cirrhosis survival analysis

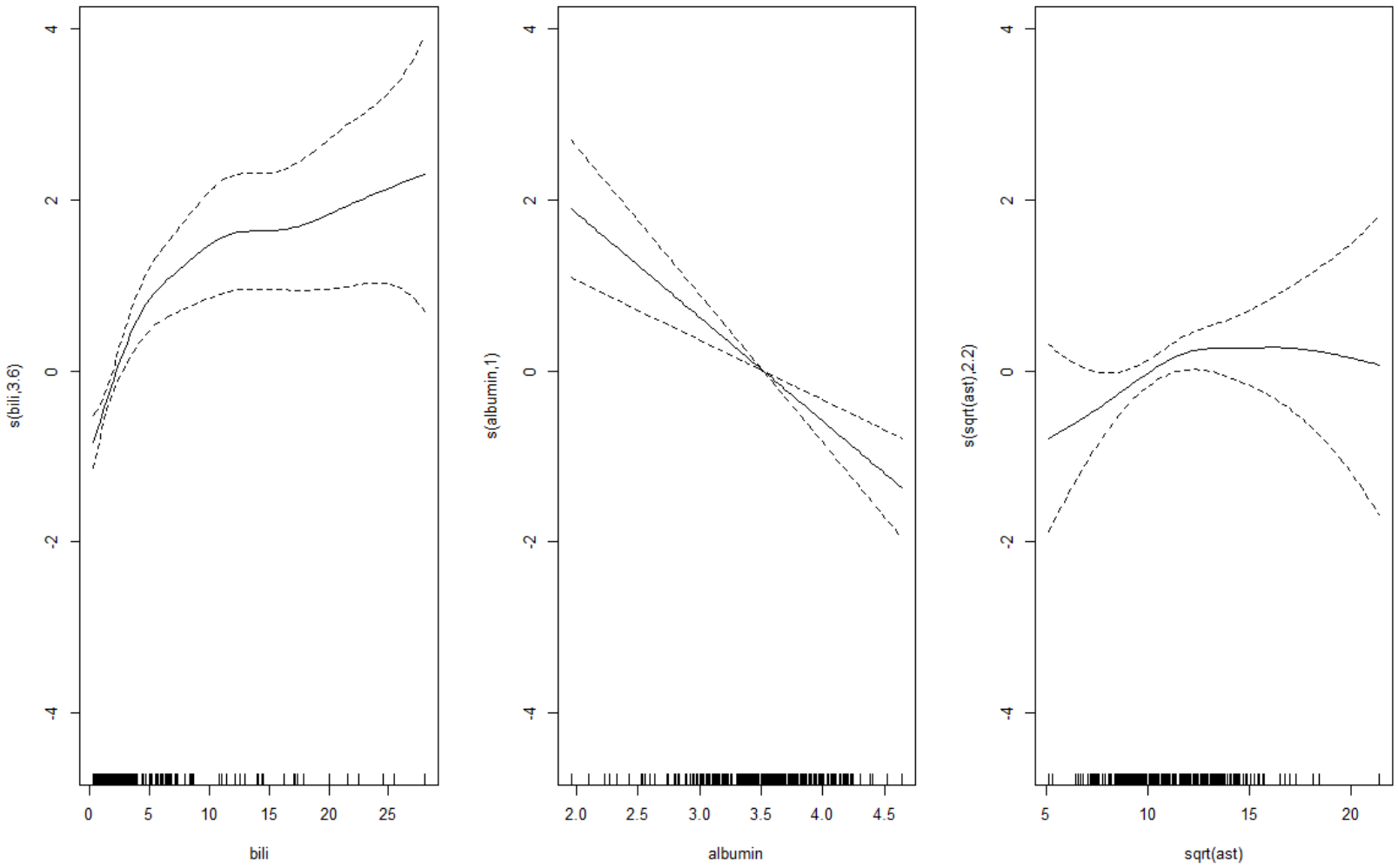
Fitted a model includes all the preceding covariates, with age and the blood test measurements included as additive smooth effects:

```
> data(pbc, package="survival")
# people who passed away at the end of the trial
> pbc$status1 <- as.numeric(pbc$status==2)
> pbc$stage <- factor(pbc$stage)
> ex3 <- gam(time ~ trt+sex+stage+s(sqrt(protime))
            + s(platelet) + s(age) + s(bili)+s(albumin)
            + s(sqrt(ast)) + s(alk.phos),
            weights=status1,
            family=cox.ph,data=pbc)
> plot(ex3)
> ex3
> summary(ex3)
```

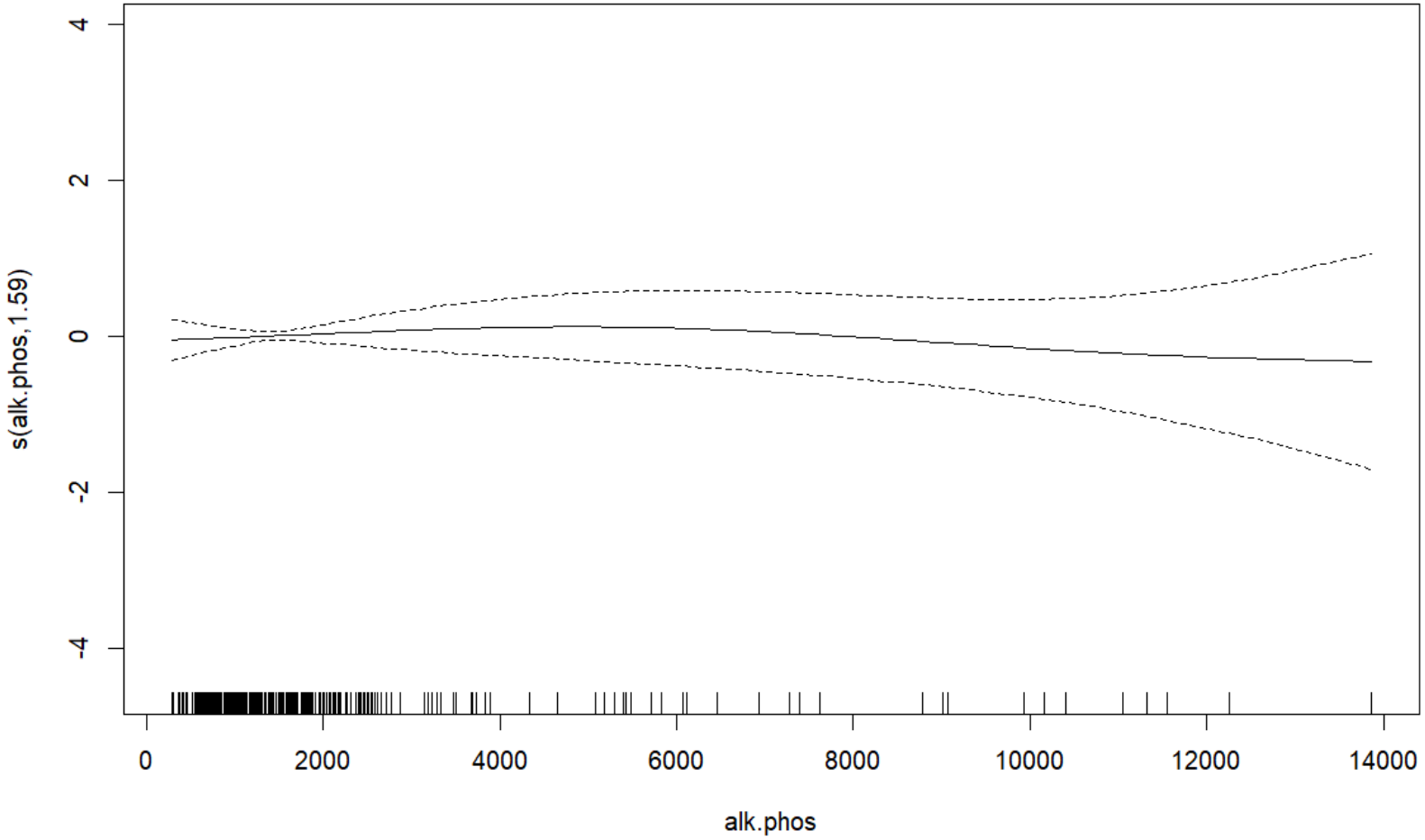
Example 3: Primary biliary cirrhosis survival analysis



Example 3: Primary biliary cirrhosis survival analysis



Example 3: Primary biliary cirrhosis survival analysis



Example 3: Primary biliary cirrhosis survival analysis

```
Family: Cox PH
Link function: identity

Formula:
time ~ trt + sex + stage + ...

Estimated degrees of freedom:
1.00 1.00 6.12 3.60 1.00 2.20 1.59 total = 21.51

REML score: 544.515
```

Example 3: Primary biliary cirrhosis survival analysis

```
Approximate significance of smooth terms:
                   edf Ref.df Chi.sq  p-value
s(sqrt(protime)) 1.000  1.000 12.901 0.000329 ***
s(platelet)      1.000  1.001  2.594 0.107313
s(age)           6.115  7.227 30.217 0.000104 ***
s(bili)          3.605  4.430 45.270  < 2e-16 ***
s(albumin)       1.000  1.000 22.186 2.71e-06 ***
s(sqrt(ast))     2.197  2.798  4.951 0.167015
s(alk.phos)      1.592  1.957  0.837 0.672275
```

Example 3: Primary biliary cirrhosis survival analysis

Fitted model has smoothers with EDFs ≈ 1.00, and some smoothers that aren't significant

Stepwise model building:

i. removed `s(alk.phos)`

ii. removed `s(sqrt(ast)`

iii. removed `stage`

Example 3: Primary biliary cirrhosis survival analysis

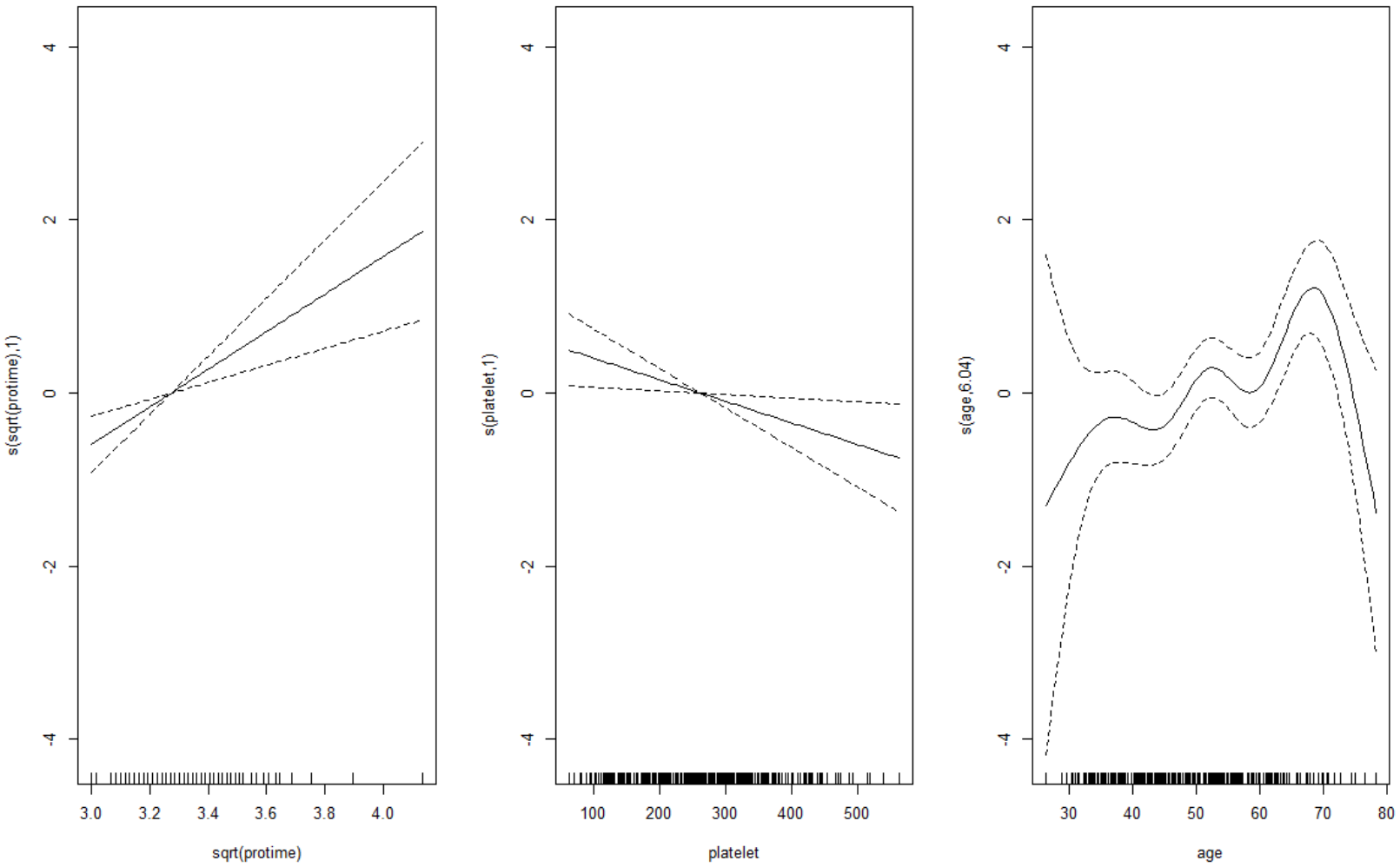
Final model:

```
> ex3_final <- gam(time ~ trt+sex+s(sqrt(protime))
                     + s(platelet) + s(age)+s(bili)
                     + s(albumin), weights=status1,
                     family=cox.ph,data=pbc)

> ex3_final
> plot(ex3_final)
> anova(ex3_final)
```

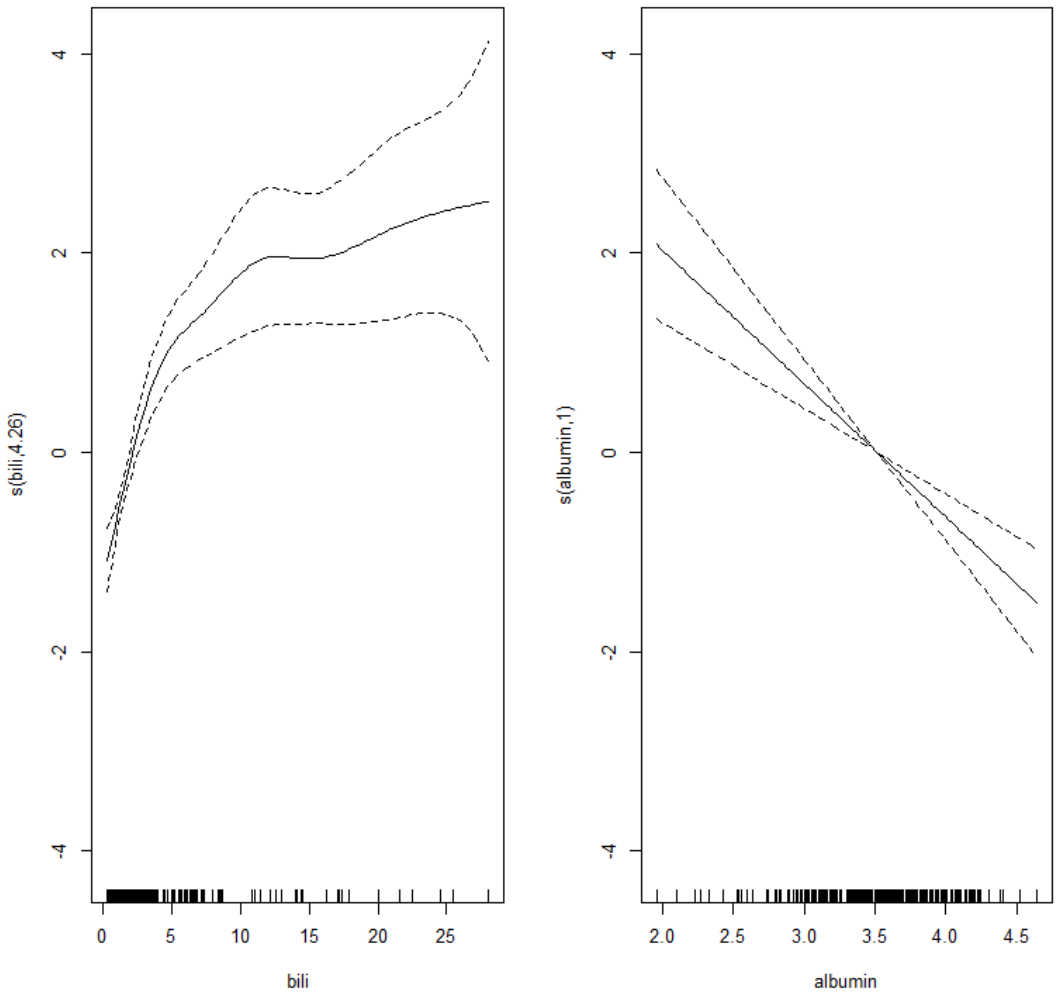
Example 3: Primary biliary cirrhosis survival analysis

```
Family: Cox PH
Link function: identity

Formula:
time ~ trt + sex + s(sqrt(protime)) + s(platelet)
 + s(age) + s(bili) + s(albumin)

Estimated degrees of freedom:
1.00 1.00 6.04 4.26 1.00  total = 15.31

REML score: 547.2567
```

Example 3: Primary biliary cirrhosis survival analysis



Example 3: Primary biliary cirrhosis survival analysis



Example 3: Primary biliary cirrhosis survival analysis

```
Family: Cox PH
Link function: identity

Parametric Terms:
    df Chi.sq p-value
trt  1  0.120  0.7294
sex  1  3.439  0.0637

Approximate significance of smooth terms:
                   edf Ref.df Chi.sq  p-value
s(sqrt(protime)) 1.000  1.001 13.337 0.000261
s(platelet)      1.001  1.002  5.789 0.016142
s(age)           6.043  7.172 29.417 0.000140
s(bili)          4.264  5.223 89.545  < 2e-16
s(albumin)       1.000  1.000 31.086  < 2e-16
```

Example 3: Primary biliary cirrhosis survival analysis

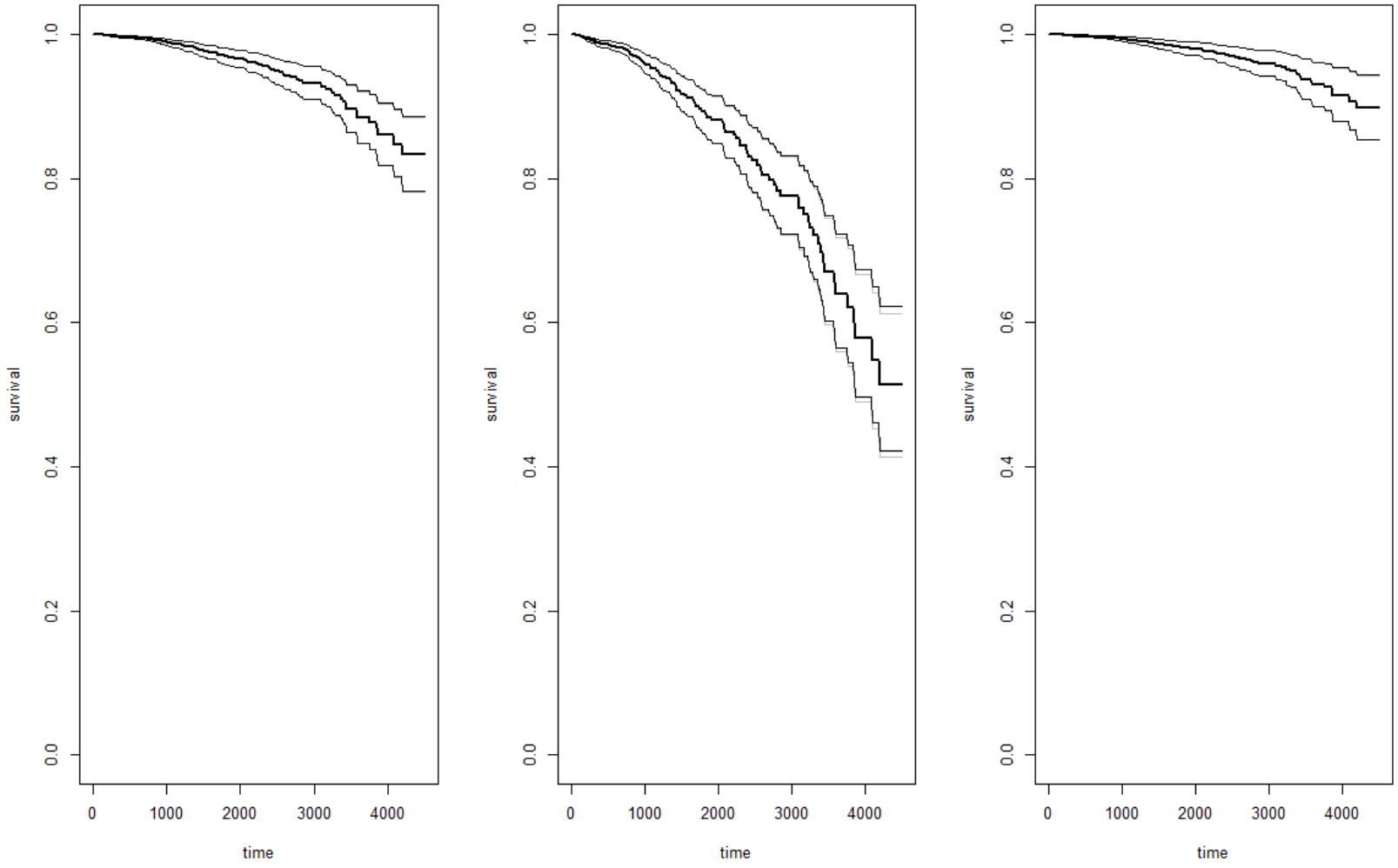


Figure: Predicted survival curves and standard error bands for three patients in the trial

Example 4: Fuel efficiency of cars

Example 4: Fuel efficiency of cars

We will use the `mpg` data from the `gamair`-library. The data were collected from import specifications, auto manuals and insurance reports.

Following information will be used to model fuel efficiency in the city and on the highway:

- `city.mpg`: city miles per gallon
- `hw.mpg`: highway miles per gallon
- `style`: style of the car (5)
- `make`: name of car maker (22)
- `fuel`: 2-level factor (gas or diesel)
- `drive`: 3-level factor (front, rear or all wheel drive)
- `weight`: in pounds
- `hp`: horse power

Example 4: Fuel efficiency of cars

Model for city and for highway fuel efficiency following model (separately):

$$\begin{aligned} mpg_i &= \beta_0 + \beta_1 fuel_i + \beta_2 style_i + \beta_3 drive_i \\ &+ f_1(weight_i) + f_2(hp_i) + f_3(make_i) + \varepsilon_i \end{aligned}$$

For $f_3(make)$, we will use random effects (`bs="re"`). The parametric term `make` is penalized by a ridge penalty.

Example 4: Fuel efficiency of cars

```
> data(mpg, package="gamair")
> ex4City <- gam(city.mpg~fuel+style+drive
+ s(weight)+s(hp)+s(make,bs="re"), data=mpg)
> ex4City
> summary(ex4City)
> plot(ex4City)
>
>
> ex4Hw <- gam(hw.mpg~fuel+style+drive
+ s(weight)+s(hp)+s(make,bs="re"), data=mpg)
> ex4Hw
> summary(ex4Hw)
> plot(ex4Hw)
```

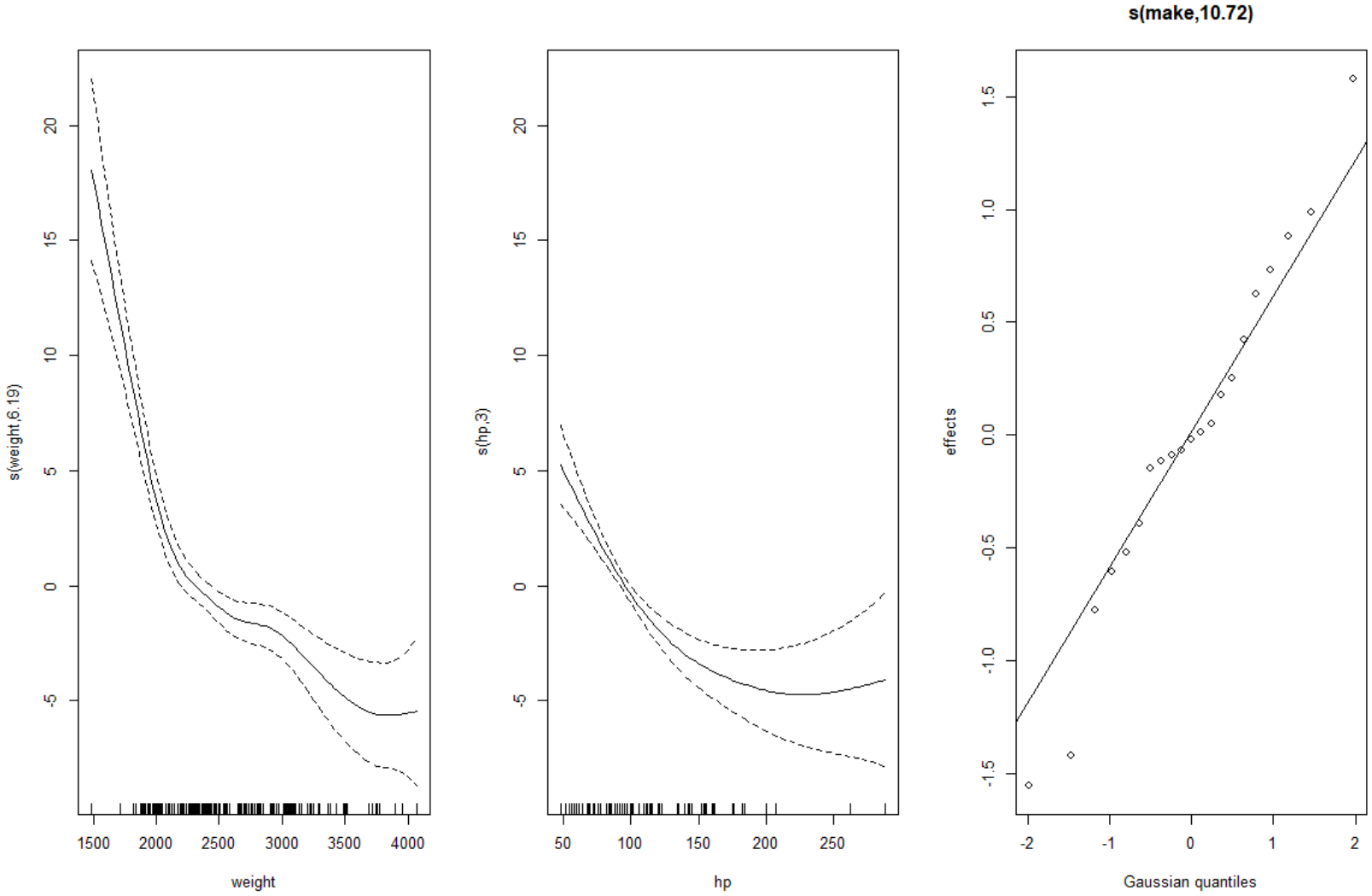
Example 4: Fuel efficiency of cars

```
Family: gaussian
Link function: identity

Formula:
city.mpg ~ fuel + style + drive +
    s(weight) + s(hp) + s(make, bs = "re")

Estimated degrees of freedom:
 6.19  3.00 10.72  total = 27.91

GCV score: 4.564226
```

Example 4: Fuel efficiency of cars

```
Approximate significance of smooth terms:
             edf Ref.df      F  p-value
s(weight)  6.195  7.279 17.504  < 2e-16 ***
s(hp)      2.999  3.707 13.132  < 2e-16 ***
s(make)   10.720 20.000  2.373 4.29e-07 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

R-sq.(adj) =  0.909   Deviance explained = 92.1%
GCV = 4.5642  Scale est. = 3.9366    n = 203
```

Example 4: Fuel efficiency of cars



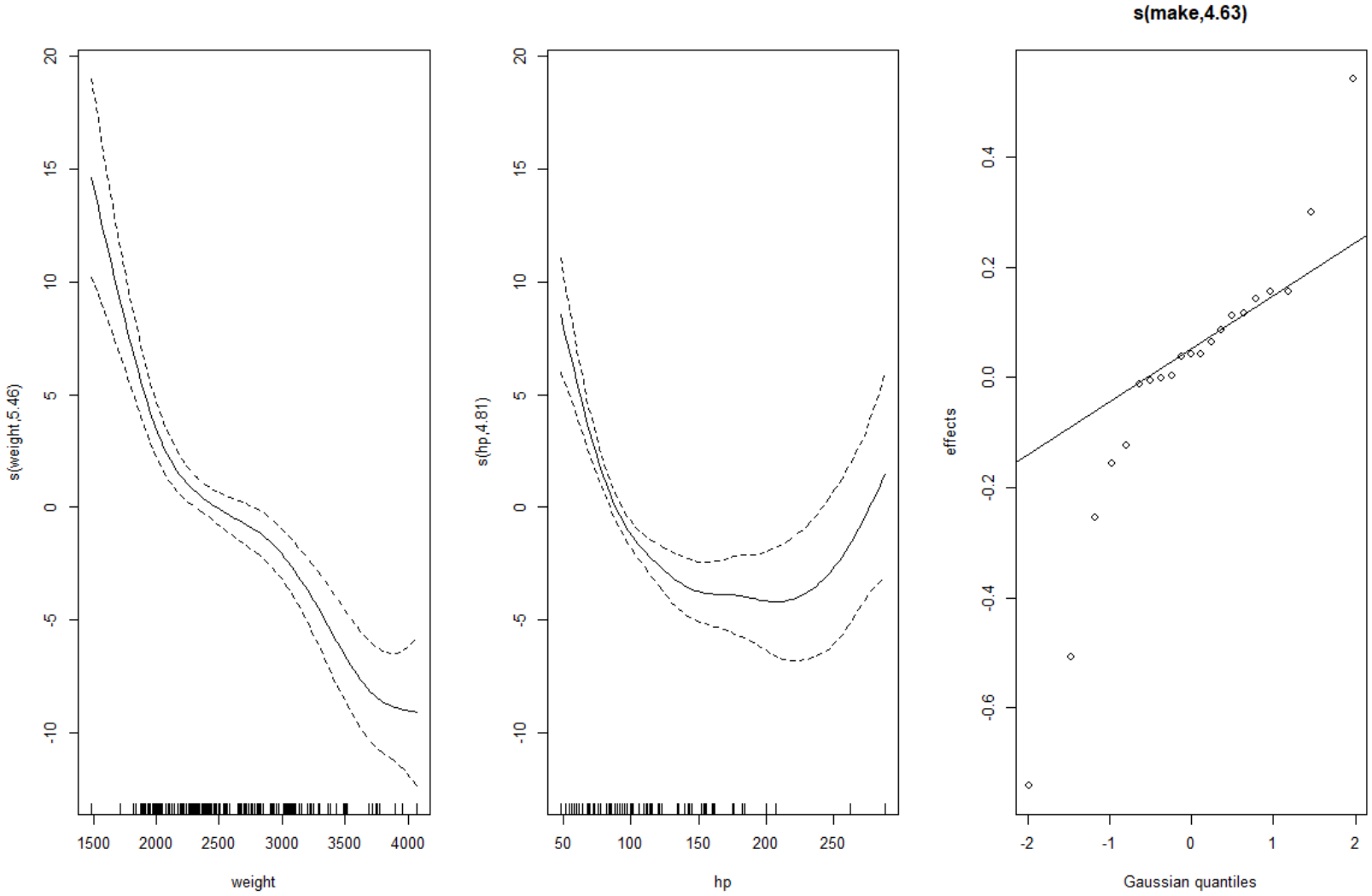
Example 4: Fuel efficiency of cars

```
Family: gaussian
Link function: identity

Formula:
hw.mpg ~ fuel + style + drive + s(weight) + s(hp) + s(make,

Estimated degrees of freedom:
5.46 4.81 4.63  total = 22.9

GCV score: 6.313085
```

Example 4: Fuel efficiency of cars

```
Approximate significance of smooth terms:
            edf Ref.df      F p-value
s(weight) 5.463  6.582 12.905  <2e-16 ***
s(hp)     4.813  5.803 11.677  <2e-16 ***
s(make)   4.626 20.000  0.432  0.0248 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

R-sq.(adj) =  0.883   Deviance explained = 89.6%
GCV = 6.3131  Scale est. = 5.6009    n = 203
```

Example 4: Fuel efficiency of cars



Example 4: Fuel efficiency of cars

It makes sense to model city fuel efficiency and highway fuel efficiency together in a bivariate normal model in which the mean of each component of the multivariate response has its own linear predictor, and there is the possibility to share terms between linear predictors. The correlations between the response components are also estimated (but do not depend on covariates).

Example 4: Fuel efficiency of cars

```
> ex4.bv <- gam(list(city.mpg~fuel+style+drive+
                     s(weight)+s(hp)+s(make,bs="re"),
                     hw.mpg~fuel+style+drive+
                     s(weight)+s(hp)+s(make,bs="re")),
                     family=mvn(d=2),data=mpg)
> ex4.bv
> summary(ex4.bv)
> plot(ex4.bv)
```

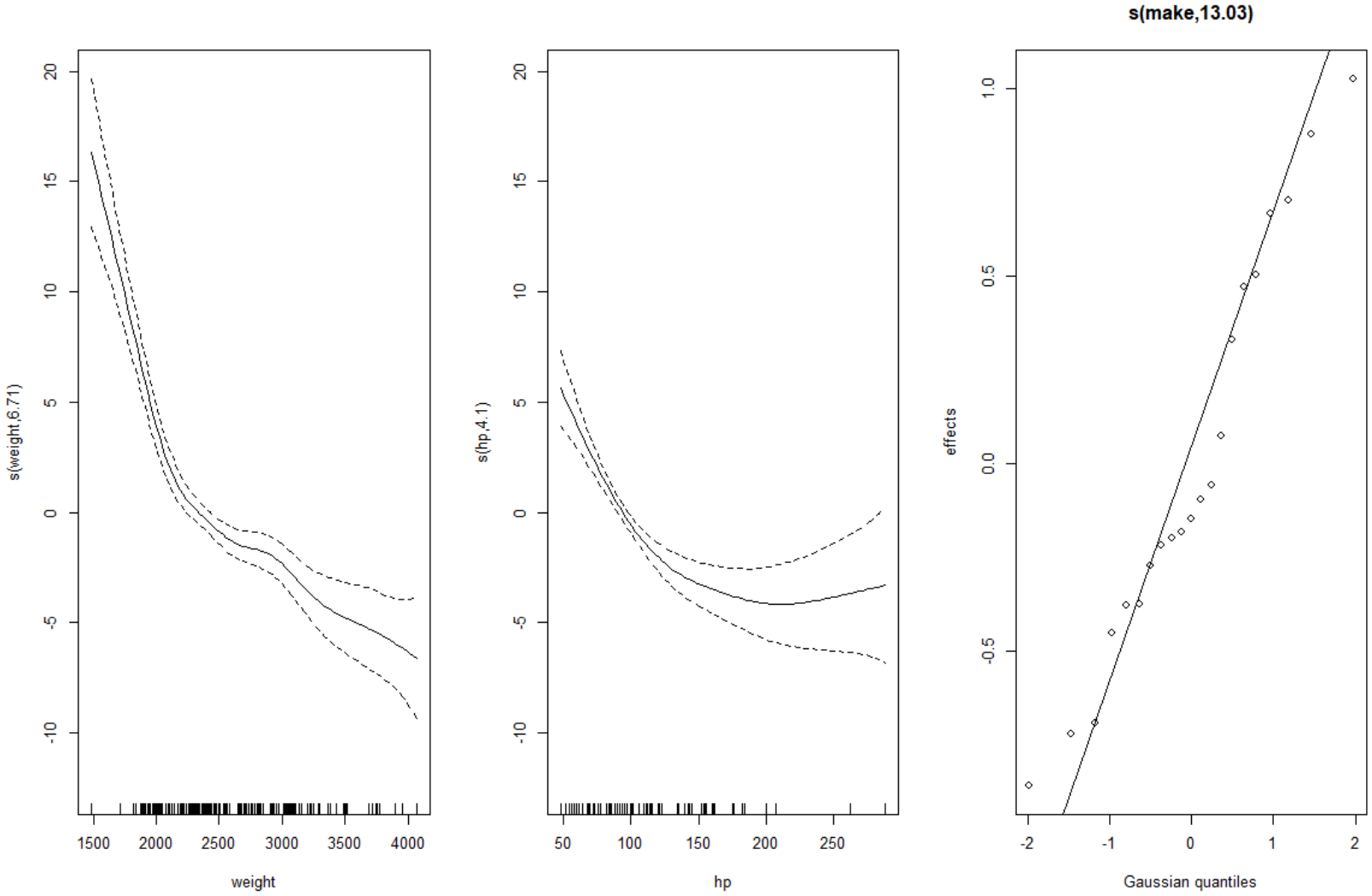
Example 4: Fuel efficiency of cars

```
Family: Multivariate normal
Link function:

Formula:
city.mpg ~ fuel + style + drive + s(weight) + ...
    bs = "re")
hw.mpg ~ fuel + style + drive + +s(weight) + ...
    bs = "re")

Estimated degrees of freedom:
 6.713  4.103 13.028  5.714  5.692  0.001  total = 54.25

REML score: 431.403
```

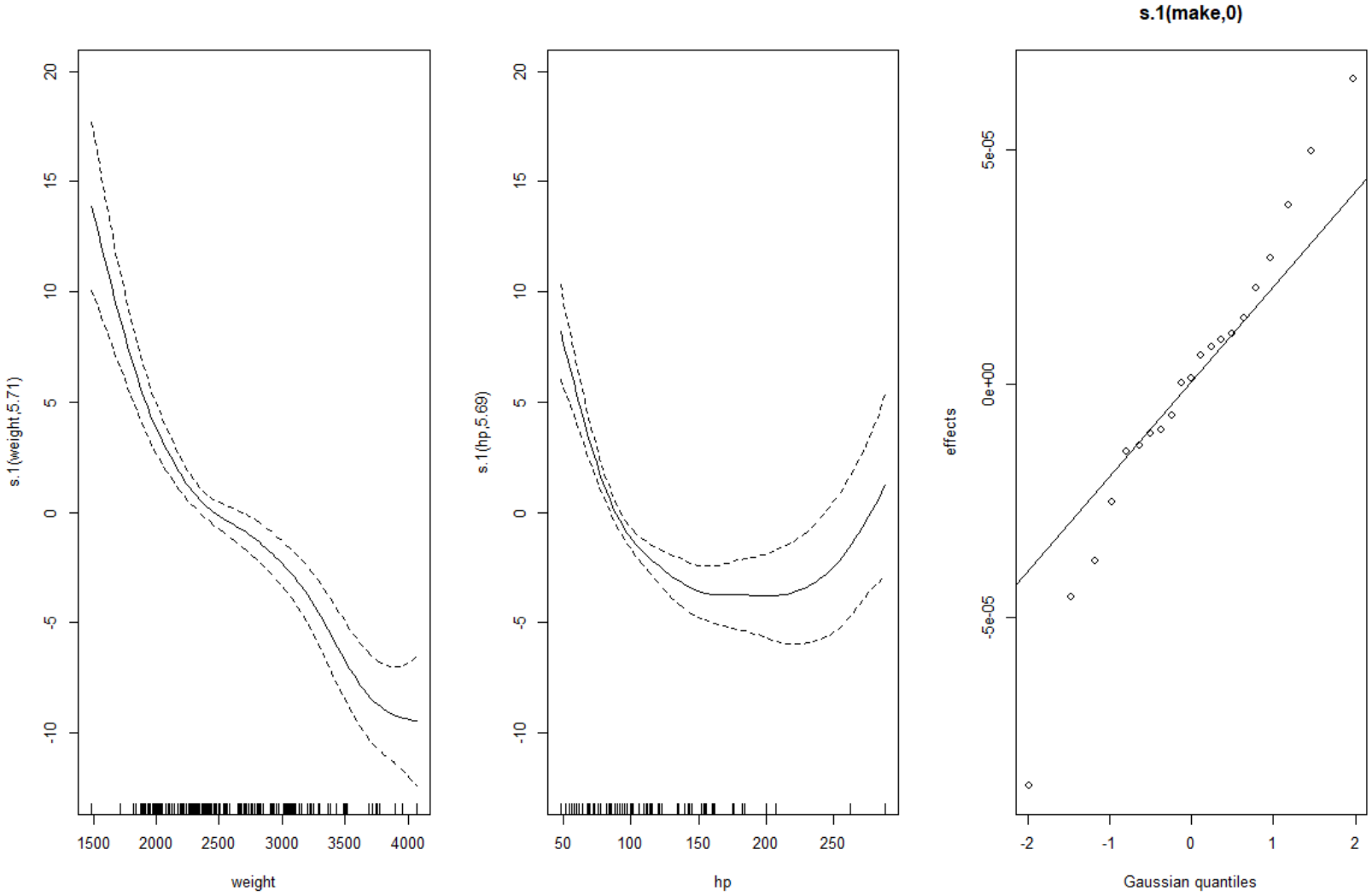
Example 4: Fuel efficiency of cars

```
Approximate significance of smooth terms:
                  edf Ref.df  Chi.sq p-value
s(weight)    6.712797  7.597 140.887  <2e-16 ***
s(hp)        4.103247  4.950  57.568  <2e-16 ***
s(make)     13.028036 20.000  71.056  <2e-16 ***
s.1(weight)  5.713765  6.739 100.066  <2e-16 ***
s.1(hp)      5.692189  6.690  77.307  <2e-16 ***
s.1(make)    0.001007 20.000   0.001   0.477
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

Deviance explained = 85.3%
-REML =  431.4  Scale est. = 1         n = 203
```

Example 4: Fuel efficiency of cars



Example 4: Fuel efficiency of cars



Example 4: Fuel efficiency of cars

Conclusions:

- Random effect for `make`: zero for highway efficiency, while clearly important for city efficiency
- Smoothers for `weight` and for `hp` are similar for both efficiencies

$\Rightarrow$ fit a new model with the same smooths of `weight` and`hp` for both `mpg` measurements, but with an extra smooth of both for the `city.mpg`. If the joint smooths are adequate then the extra smooths should turn out to be unnecessary.

Example 4: Fuel efficiency of cars

```
> ex4.bv2 <-  gam(list(city.mpg ~ fuel + style +drive
                 + s(hp) + s(weight)
                 + s(make, bs="re"),
                 hw.mpg ~ fuel + style + drive
                 + s(make,bs="re"),
                 1+2 ~ s(weight) +s(hp) -1),
                 family = mvn(d=2) , data = mpg)
> ex4.bv2
> summary(ex4.bv2)
> plot(ex4.bv2)
```

Example 4: Fuel efficiency of cars

```
Family: Multivariate normal
Link function:

Formula:
city.mpg ~ fuel + style + drive + s(hp) + ...

Estimated degrees of freedom:
 4.2510  4.2873 13.0362  0.0014  5.7493  4.9389

total = 51.26

REML score: 427.3227
```

Example 4: Fuel efficiency of cars

```
Approximate significance of smooth terms:
                  edf Ref.df  Chi.sq  p-value
s(hp)        4.250975  5.141  33.657 4.16e-06 ***
s(weight)    4.287317  5.221  43.926  < 2e-16 ***
s(make)     13.036244 20.000  62.536  < 2e-16 ***
s.1(make)    0.001424 20.000   0.002    0.243
s.2(weight)  5.749315  6.909 105.539  < 2e-16 ***
s.2(hp)      4.938945  5.961  75.711  < 2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘

Deviance explained = 85.2%
-REML = 427.32  Scale est. = 1         n = 203
```

Example 4: Fuel efficiency of cars

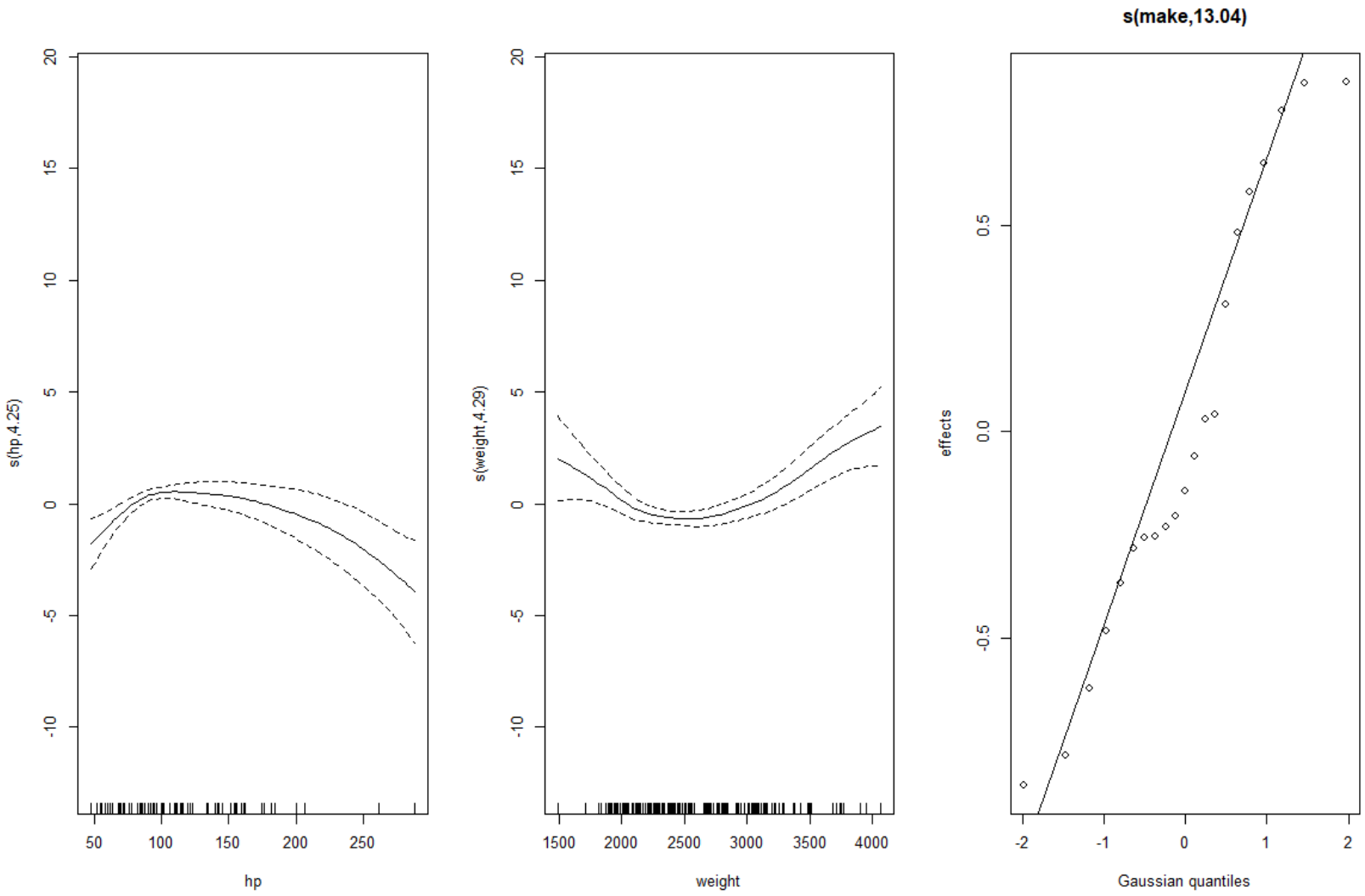


Figure: Results for city mpg

Example 4: Fuel efficiency of cars

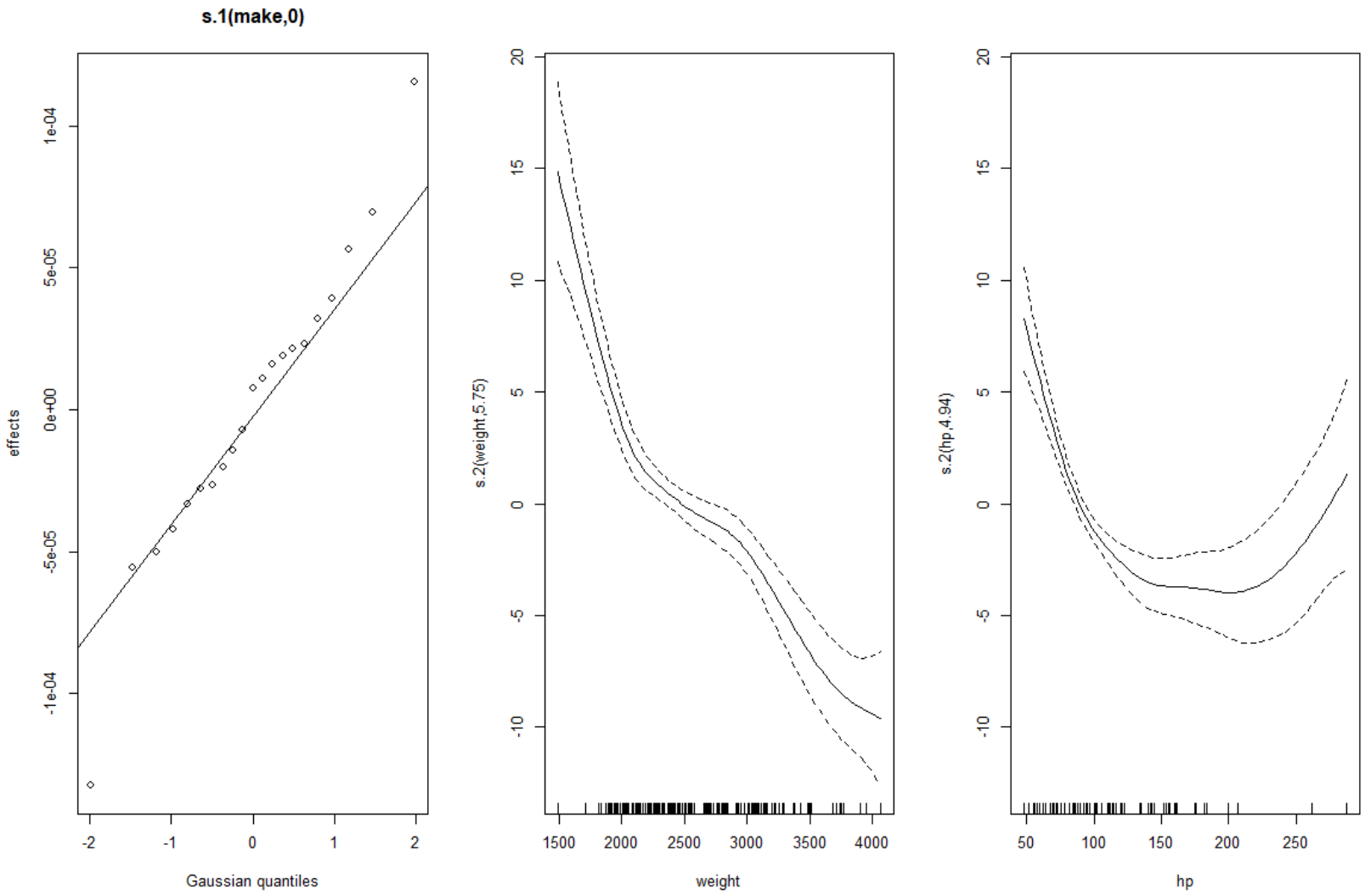


Figure: Results for highway mpg